

Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 7 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 7 : SLO – 1 | Data Structures — Definition and Types

■ I. Fill in the Blanks

1. data structure — a way of organizing and storing data for efficient access and modification.
2. linear and non-linear — the two broad types.
3. linear — elements arranged sequentially.
4. Queue — the FIFO linear data structure.
5. Tree — the non-linear structure for hierarchical relationships.
6. graph — elements point to multiple elements, forming a network.
7. push and pop — the two stack operations.
8. array — a collection of elements identified by index.
9. linked list — nodes containing data plus a next pointer.
10. storage (organization) and retrieval (processing) — efficient management of data.

■ II. Match the Following

1. Array — C. Sequential memory allocation.
2. Stack — A. Last In First Out.
3. Queue — B. First In First Out.
4. Linked List — D. Each node points to the next node.
5. Tree — J. Hierarchical structure.
6. Graph — H. Represents complex relationships.
7. Linear Data Structure — F. Stack (example).
8. Non-Linear Data Structure — I. Graph (example).
9. Stack Operations — G. Push and Pop.
10. Queue Operations — E. Enqueue and Dequeue.